

Prelab 6: Slow Decay (RC Circuits)

March 13, 2026

1. Perform a Kirchhoff analysis of the following RC circuit (Resistor-Capacitor circuit) and determine an equation for the current in the circuit as a function of time, $i(t)$. Calculate the time constant for this circuit.

Using KVL:

$$\mathcal{E} - iR - \frac{Q}{C} = 0.$$

Since $i = dQ/dt$, this becomes a first-order ODE:

$$\mathcal{E} - R \frac{dQ}{dt} - \frac{Q}{C} = 0 \Rightarrow \frac{dQ}{dt} = \frac{C\mathcal{E} - Q}{RC}.$$

Separating variables and integrating (with $Q(0) = 0$):

$$\int_0^Q \frac{dQ'}{C\mathcal{E} - Q'} = \int_0^t \frac{dt'}{RC} \Rightarrow Q(t) = C\mathcal{E}(1 - e^{-t/RC}).$$

Differentiating to get the current:

$$i(t) = \frac{dQ}{dt} = \frac{\mathcal{E}}{R} e^{-t/RC}.$$

The time constant is $\tau = RC$. With $R = 22 \text{ M}\Omega$ and $C = 1 \text{ }\mu\text{F}$:

$$\tau = RC = (22 \cdot 10^6 \Omega)(1 \cdot 10^{-6} \text{ F}) = 22 \text{ s}.$$

2. Write a few sentences that explain what the time constant means. Do not use any equations.

The time constant τ describes how slowly the current in the circuit decays over time. A larger time constant means the current decreases slower, while a smaller time constant means it decays faster.

3. You should already know the definition of the half-life ($\tau_{1/2}$) of a decay. What is the $1/e$ life? Derive the formula for the $1/e$ life. Why is this quantity useful?

The $1/e$ life, $\tau_{1/e}$, is the time it takes for the quantity to fall to $1/e \approx 36.8\%$ of its initial value. For the current $i(t) = i_0 e^{-t/\tau}$, we set $i(\tau_{1/e}) = i_0/e$:

$$i_0 e^{-\tau_{1/e}/\tau} = \frac{i_0}{e} = i_0 e^{-1} \Rightarrow \frac{\tau_{1/e}}{\tau} = 1 \Rightarrow \tau_{1/e} = \tau = RC.$$

The $1/e$ life is useful because it equals the time constant τ directly, so it is more intuitive to calculate and interpret.

